

for loop:-

for (initialisation; condition; updation)

Swapping Two numbers:-

$a=10, b=20 \xrightarrow{\text{after swap}} a=20, b=10$

Ist way

a $\boxed{10}$
20

b $\boxed{20}$
10

temp $\boxed{10}$

int temp = a;
a = b;
b = temp;

IInd way

$a = a + b$
 $b = a - b$
 $a = a - b$

a
 $\boxed{10}$

b
 $\boxed{20}$

$\boxed{30}$

$\boxed{10}$

$\boxed{20}$

$b = a - b$
 $= a + \cancel{b} - \cancel{b} = a$

$$a = a * b;$$

$$b = a / b;$$

$$a = a / b;$$

$$[-2^{31}, 2^{31} - 1]$$

$$(36 = \underline{1}, 2, 3, 4, 6, 9, 12, 18, 36)$$

$$\text{whole } (i \leq n)$$

$\downarrow$
 $\underline{1} \times \underline{36}$ ✓
 $\underline{2} \times \underline{18}$ ✓
 $\underline{3} \times \underline{12}$ ✓
 $\underline{4} \times \underline{9}$ ✓
 $\underline{6} \times \underline{6}$ ✓
 $\underline{9} \times \underline{4}$ X
 $\underline{12} \times \underline{3}$ X
 $\underline{18} \times \underline{2}$ X
 $\underline{36} \times \underline{1}$ X

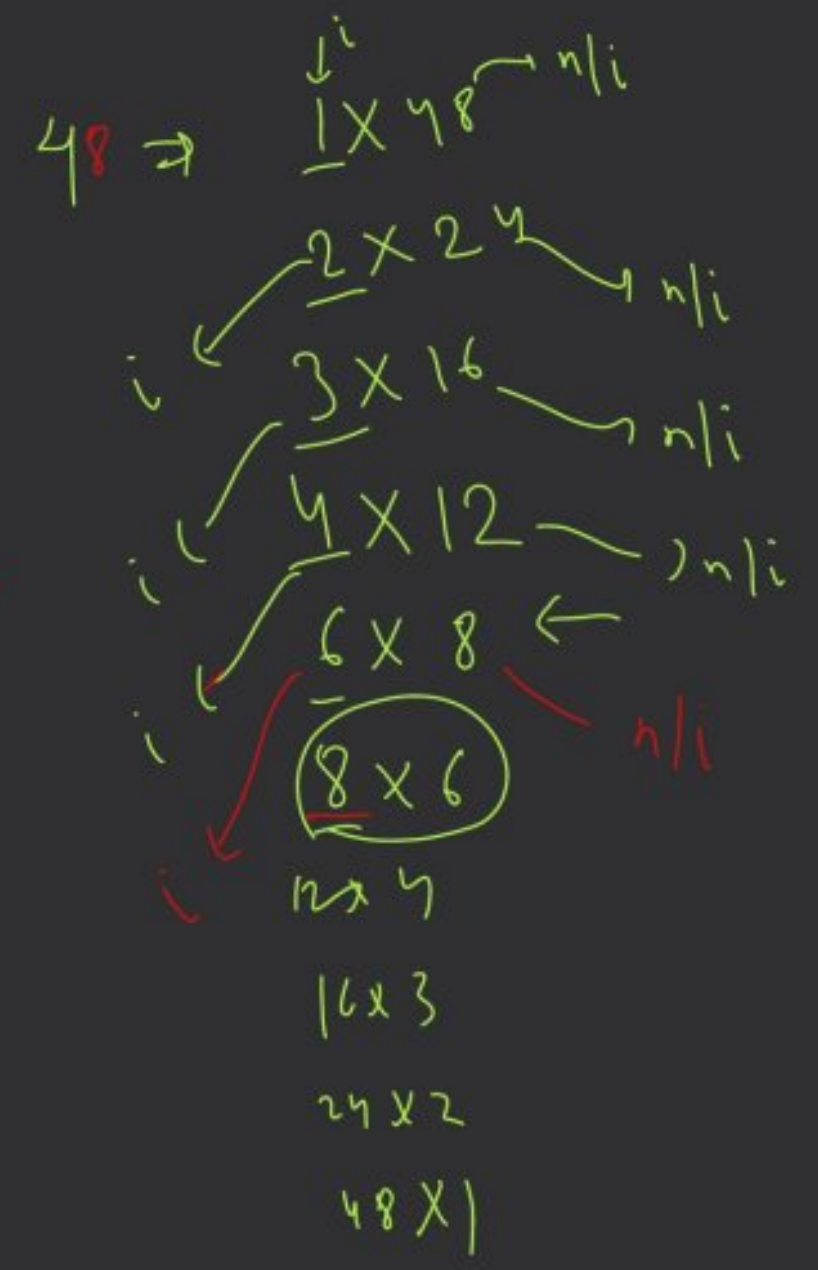
$\frac{36}{6}$
 $\frac{36}{2}$
 $\frac{36}{3}$
 $\frac{36}{4}$

```

int n=48;
while (i*i <= n) {
    if (n%i == 0) {
        cout << i;
        cout << " " << n/i;
        i++;
    }
}

```

(8×8)



64 → 1 × 64

2 × 32

4 × 16

8 × 8

16 × 4

32 × 2

64 × 1

1, 64, 2, 32, 4, 16, 8

$i \times i = 1 \times 16 \times 64$

$i = 8$

fl. W // Print Prime no. with T.C.


```
for(int i=1; i<=100; i++) {
```

```
    print(i);
```

```
}
```

```
    if (i==10 || i==20 || i==30 || i==40 || i==50)
```

```
        continue;
```

```
}
```

```
if(i==2) continue;
```

(10, 20, 30, 40, 50)

continue;

if ()

continue; or break;



We will skip
that element



We will skip the
whole iteration (i.e. We will

the loop)

```
for (i = 1 → 100) {  
    if (i == 1)  
        break; ←  
    sort(i);  
}
```

move out from

break

for (int i=1; i<=100; i++) {

if ($i >= 2$ && $i <= 99$)

continue;

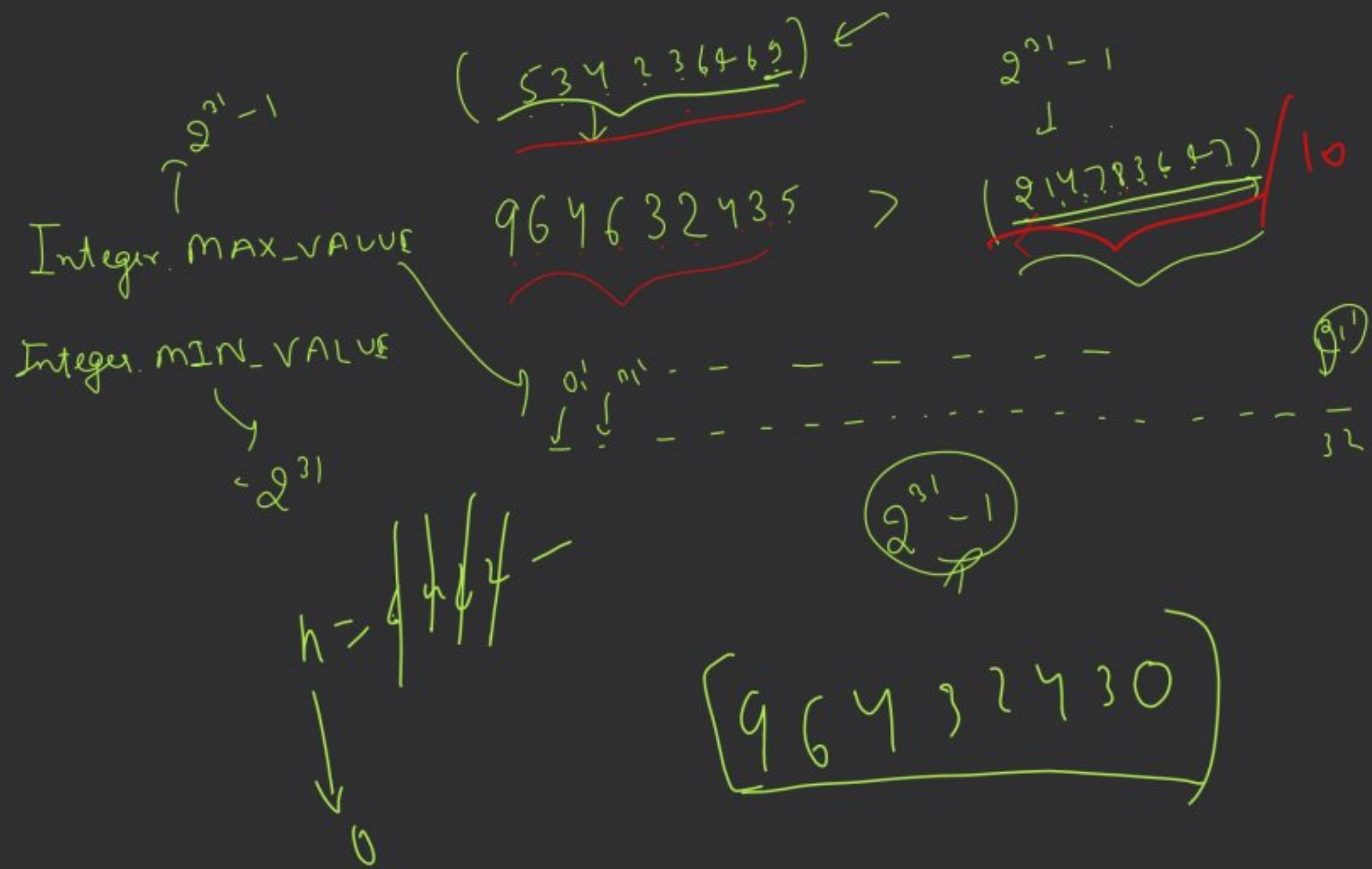
print(i);

}

(100)

1
2
}
{
1
1
1
1
100

2
99



```

while (n != 0) {
    int digit = n % 10;
    revNo = (revNo * 10) + digit;
    n = n / 10;
}
return revNo;

```

$(2^{31}-1) \times 10$

$(revNo > INT_MAX / 10)$

$\uparrow \quad \uparrow$

if $(revNo > INT_MAX / 10)$

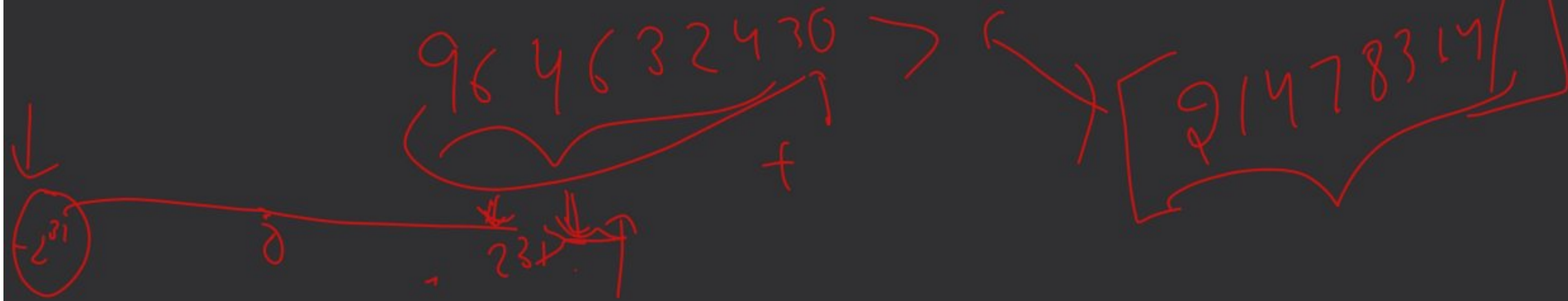
$\quad \quad \quad \uparrow$

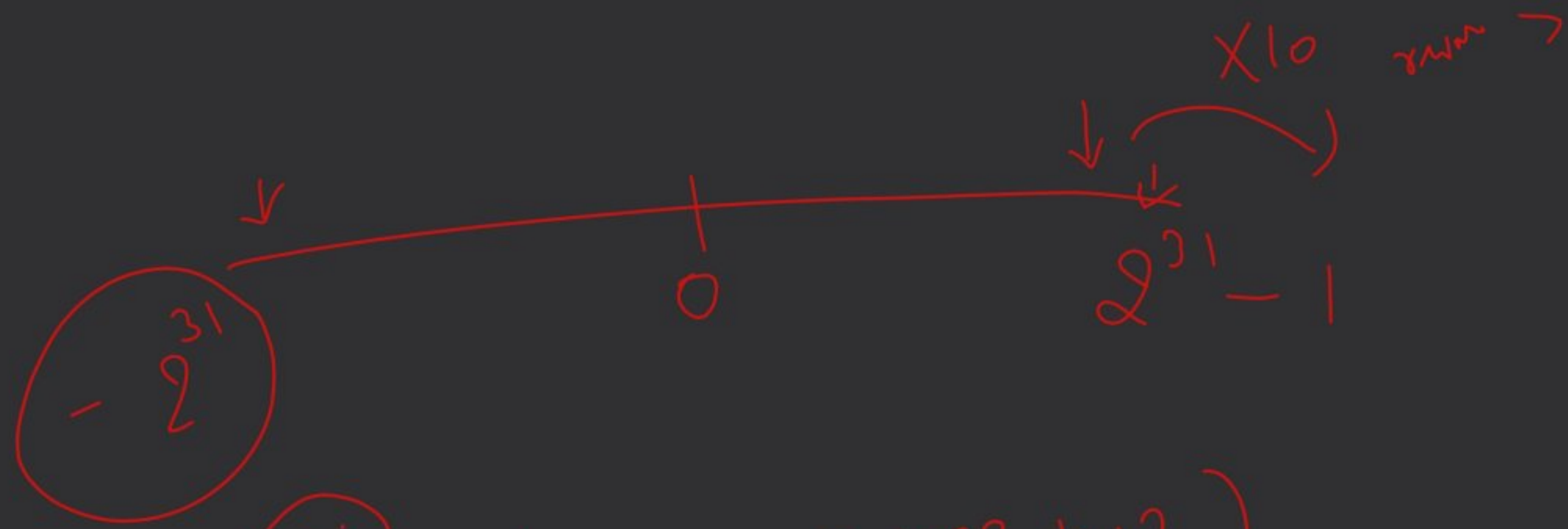
$\quad \quad \quad \text{MAX_VALUE} / 10$

|| $revNo < \text{Integer}$

$\quad \quad \quad \uparrow$

$\quad \quad \quad \text{MIN_VALUE} / 10$





Integer.MAX_VALUE/10

Integer.MIN_VALUE/10

(H.W.)

(-2147483648)

if (revNo > Integer.MAX_VALUE/10
|| revNo < Integer.MIN_VALUE/10)

return 0;

}

A, B, C

A, B

B, C

A, C

11
C
11

(9, 9)

11

10

11!

10! (11-10)!

21
12

A, B
B, n

n!

$$nCr = \frac{n!}{r!(n-r)!}$$

10 x 9 x 8 x 7 x 6 x 5 x 4 x 3 x 2 x 1

$${}^4C_2 = \frac{4!}{2!(4-2)!} = \frac{4!}{2! \times 2!}$$

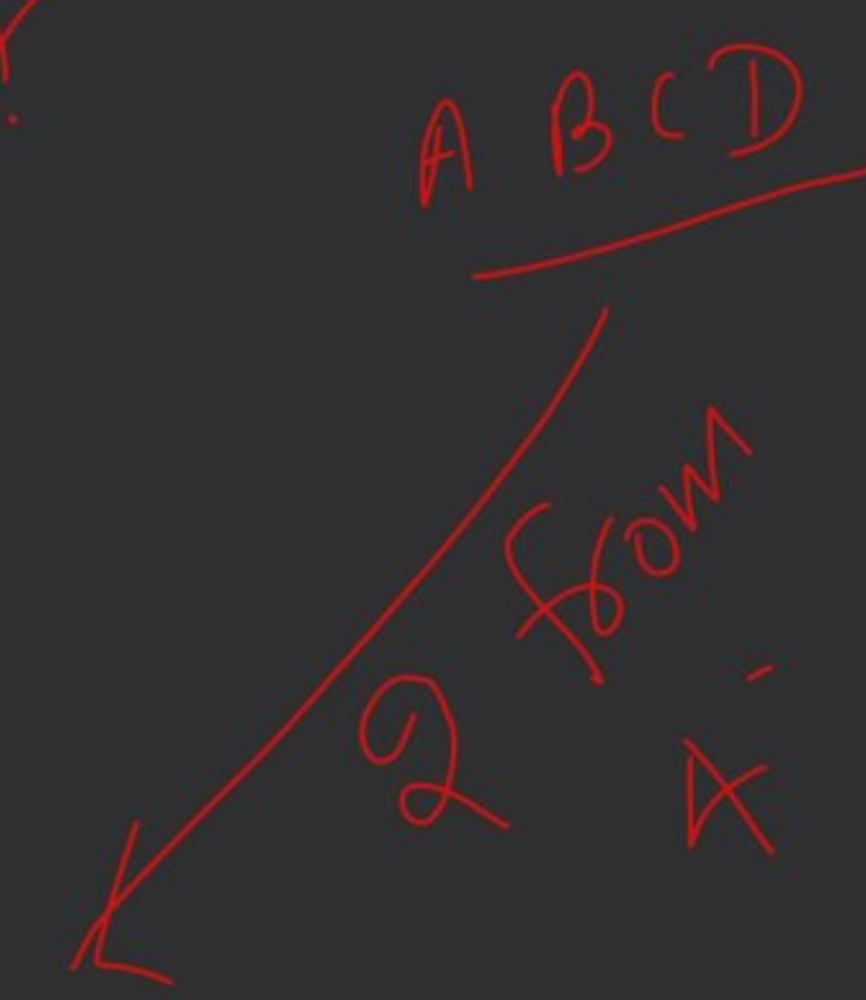
picking 2 person from a

4 person.

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$$= \frac{4 \times 3 \times \cancel{2!}}{\cancel{2!} \times \cancel{2!}} = 6$$

- AB -
- AC -
- AD -
- BC -
- BD -
- CD -



$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

$$(a+b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$$

$$(a+b)^5 = \frac{5!}{5!0!} a^5 + \frac{5!}{4!1!} a^4b + \frac{5!}{3!2!} a^3b^2 + \frac{5!}{2!3!} a^2b^3 + \frac{5!}{1!4!} ab^4 + \frac{5!}{0!5!} b^5$$

$\frac{5!}{0!5!} = \frac{5!}{1 \times 5!} = 1$

$\frac{5!}{2!3!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 3 \times 2 \times 1} = 10$



